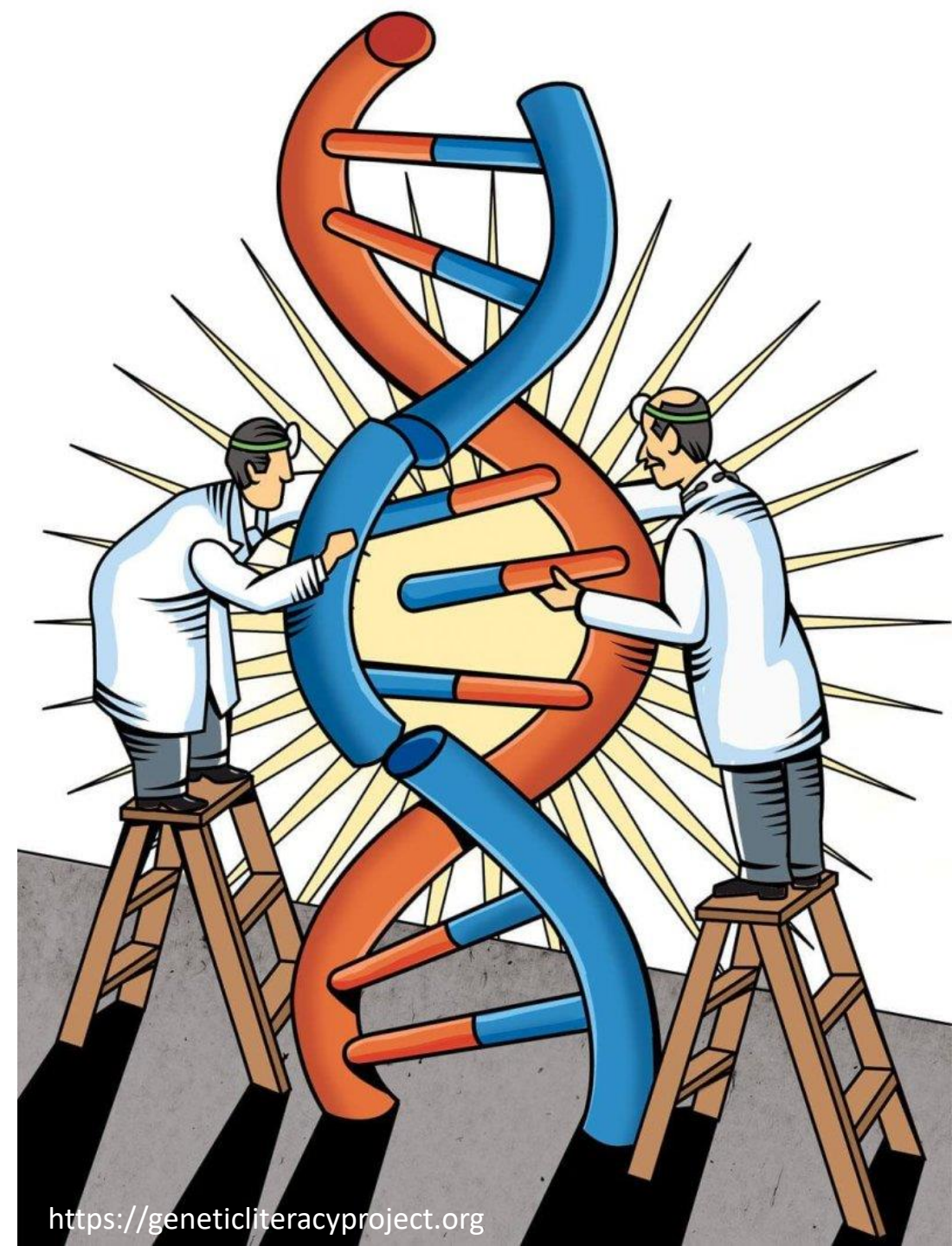


CRISPR in action

practice and theory of genome engineering

Day 4
August 20, 2026

Sibylle Vonesch
Laboratory for Genome Editing and
Systems Genetics



Genotyping approaches

PCR fragment analysis (gel/fragment size)

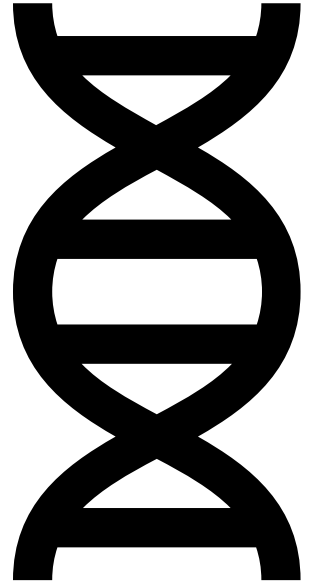
- Detects deletions/insertions by amplicon size shift

Sanger sequencing

- Confirms precise base edits or small indels
- Requires clean primer placement for good reads

Amplicon sequencing (NGS)

- Sensitive detection of mosaicism or low-frequency edits/genotypes
- Quantitative allele distribution
- Requires short amplicons (NGS read length)



PCR fragment analysis

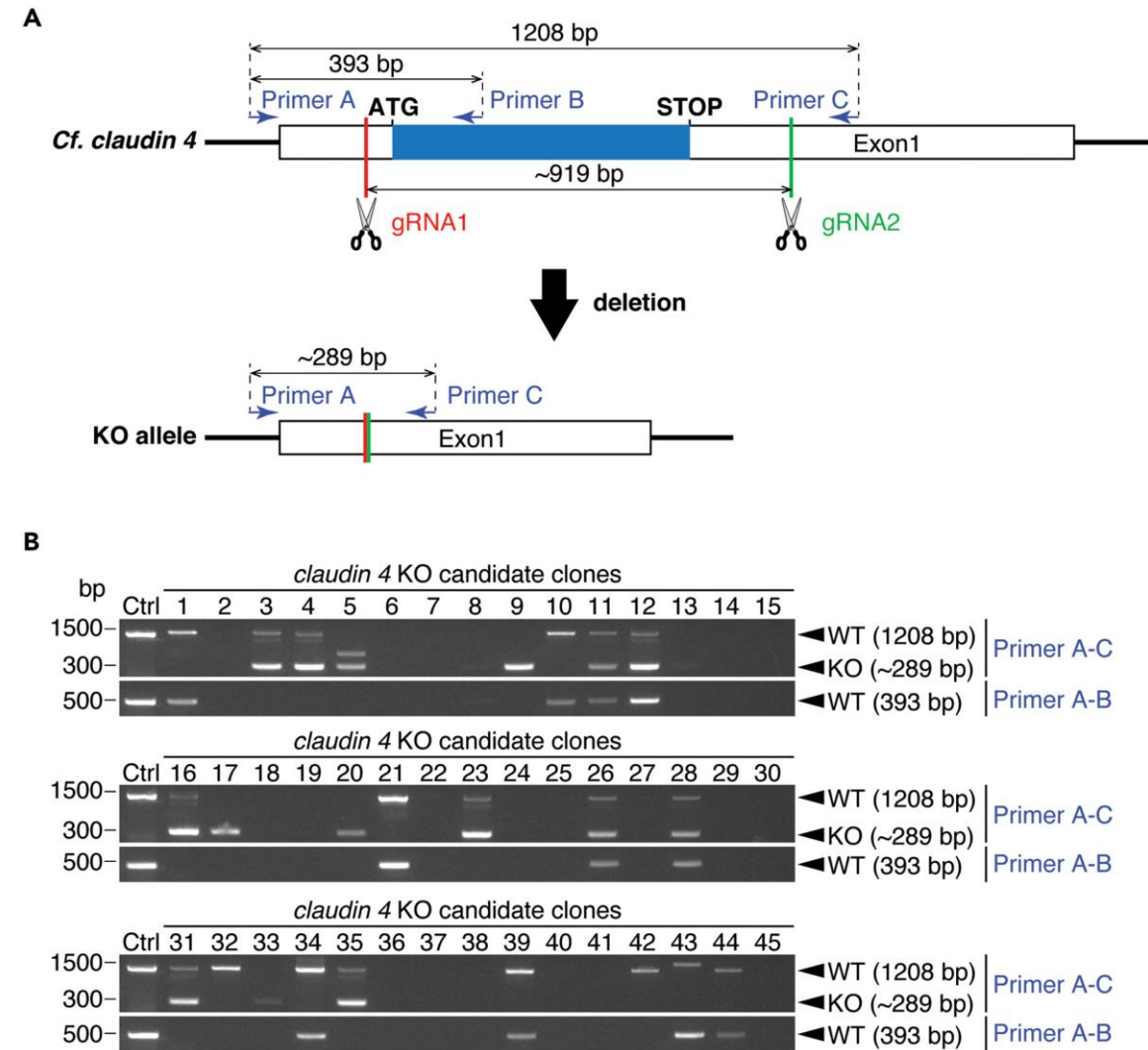
Primers should be able to amplify wild-type and edited alleles

Shorter genes/loci: one PCR, size difference

Longer genes/loci: two PCRs, presence/absence of bands

Design good PCR primers (manual, primer3 etc):

- 18–24 bp, GC 40–60%, T_m 55 - 65 °C
- Avoid dimers, hairpins, repetitive bases
- G or C at 3' end
- Check primer uniqueness (BLAST)



BLAST options

NCBI-BLAST

Organism-specific database

Primer-BLAST

<https://www.ncbi.nlm.nih.gov/tools/primer-blast/>

BLAST+ (command line)

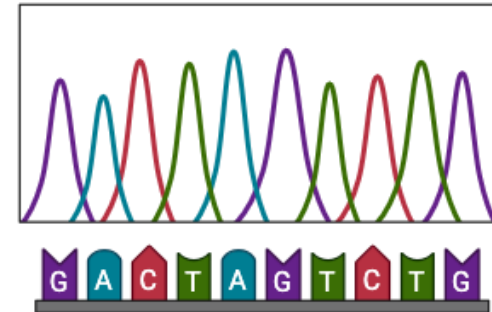
<https://conmeehan.github.io/blast+tutorial.html>

The screenshot shows the NCBI BLAST website. At the top is the NIH logo and the text "National Library of Medicine National Center for Biotechnology Information". A "Log in" button is in the top right. Below the header is a navigation bar with "BLAST®" and links for "Home", "Recent Results", "Saved Strategies", and "Help". A blue banner contains an "Important update" about the "ClusteredNR" database. The main content area features the "Basic Local Alignment Search Tool" description, a "Web BLAST" section with three buttons: "blastx" (translated nucleotide to protein), "tblastn" (protein to translated nucleotide), and "Protein BLAST" (protein to protein). A "Nucleotide BLAST" button (nucleotide to nucleotide) is also present. A "NEWS" sidebar on the right shows a date "Mon, 21 Jul 2025" and a link to "More BLAST news...".

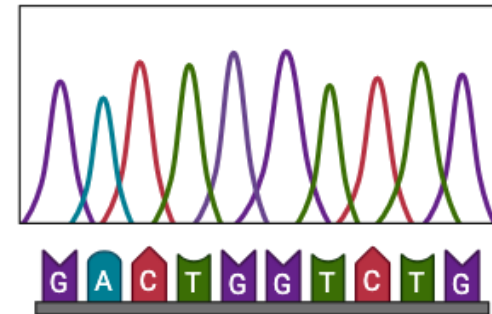
Sanger

- Short indels, precision edits
- Amplicons preferentially <900 bp length
- Edit > 30 and < 850 bp from primer
- Avoid repetitive or homopolymer runs in amplicon, if unavoidable try to avoid in sequencing (use fwd, rev primer)
- Single clean band essential

Design good PCR/sequencing primers!



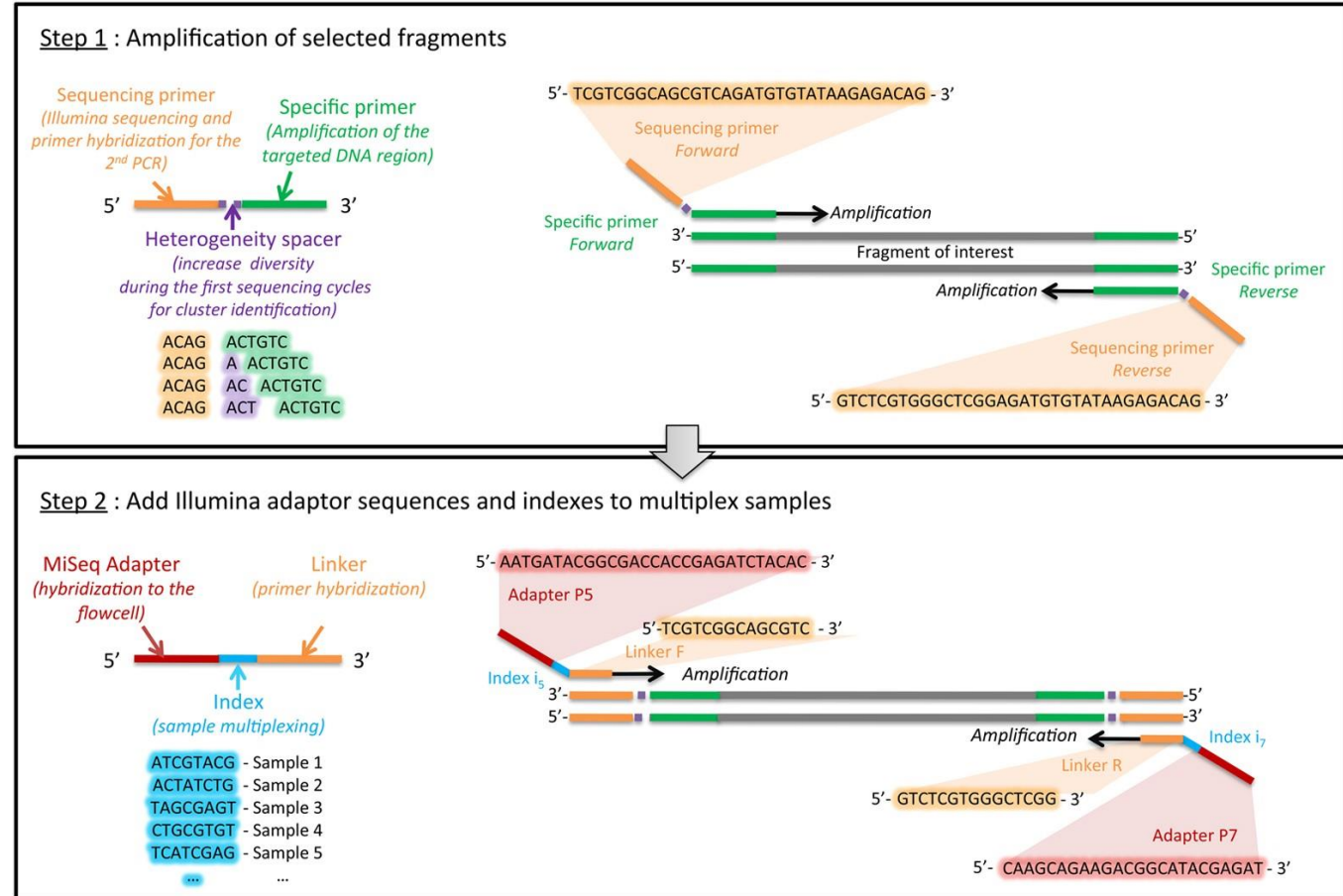
Wild-type



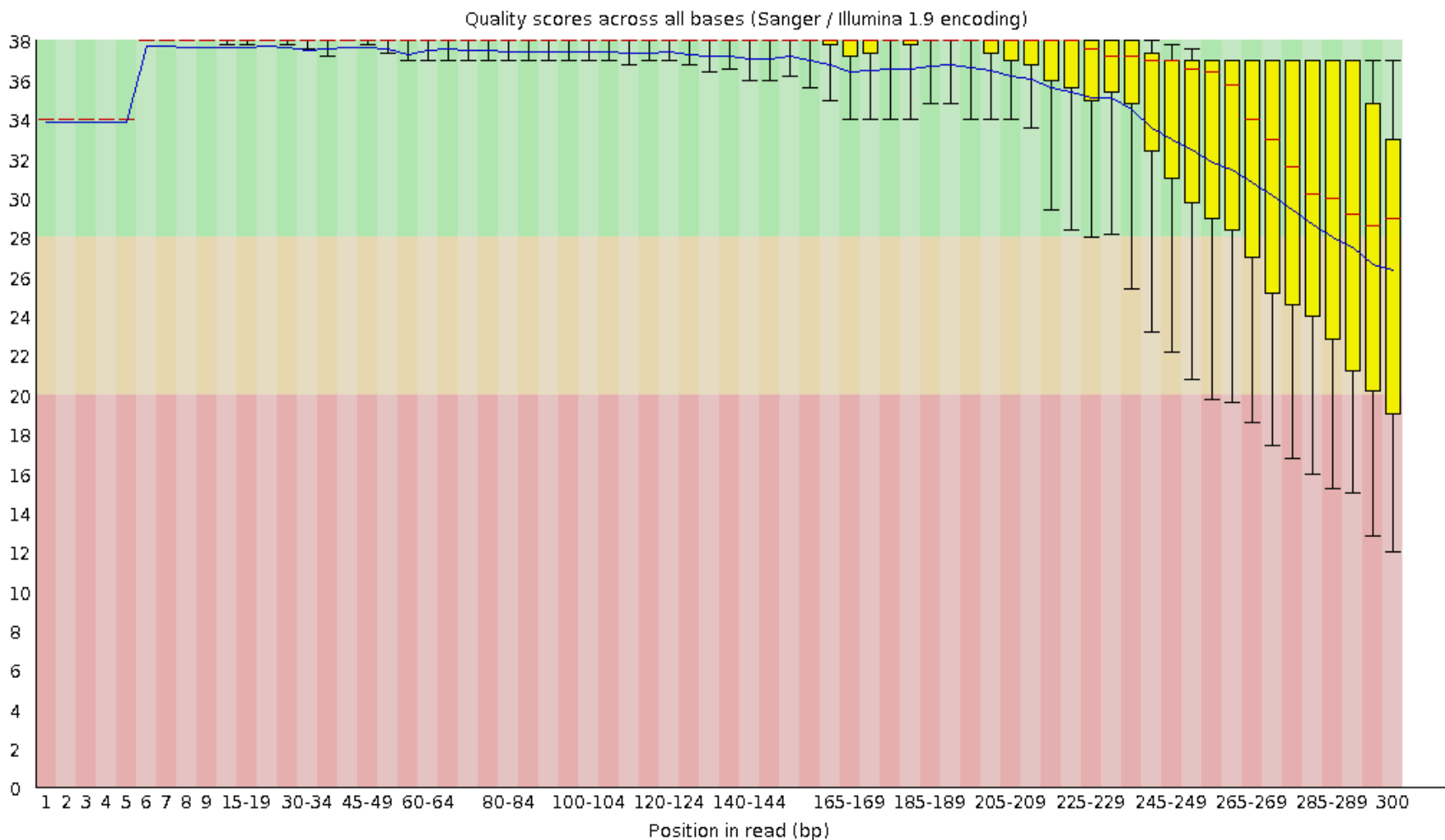
Edit (A to G)

Amplicon-seq

- Allele distributions, quantitative
- Amplicon length constrained by NGS read length
- Short-read: 150 PE ideal, amplicon 180 – 230 bp (max. 270 bp), edit in middle → consensus reads
- Multiple bands tolerated but not ideal (waste reads)
- Design good PCR/sequencing primers (primer3 + BLAST for design)
- Two PCRs: target specific primers, Illumina compatible primers + adapters



Sequence quality dropoff





Go to *galK* file on Benchling and design primers for genotyping your PTC and KO